

# Abhinandan Onajol

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## Professional Summary

Recent Computer Science (AI) graduate with a peer-reviewed publication at ACL 2024 on LoRA fine-tuning and hands-on experience shipping end-to-end AI systems. Proficient in LLM integration, multi-agent pipeline design, real-time computer vision inference, and on-device model deployment. Experienced across the full ML lifecycle: data engineering, model training, hyperparameter optimisation, model compression, and production API delivery.

## Core Competencies

LLMs — LoRA / PEFT Fine-Tuning — RAG — Multi-Agent AI Systems — Prompt Engineering — NLP — FastAPI — Computer Vision — On-Device ML

## Education

**B.E. — Computer Science (Artificial Intelligence)** — KLE Technological University, Hubballi, Karnataka *Nov 2022 – Jun 2026*  
CGPA: 8.57/10 — Relevant Coursework: Embedded Intelligent Systems, Machine Learning & Deep Learning

**Pre-University Certificate — Science Stream** — Ekalavya PU College, Belagavi, Karnataka *Aug 2020 – Mar 2022*  
Aggregate: 96.17% — Ranked 1<sup>st</sup> in college

## Professional Experience

**AI Engineering Intern** — Bitlance Tech Hub Pvt Ltd, Pune *Dec 2025 – Feb 2026*

- **Designed and deployed production-grade FastAPI backend services** using Uvicorn, building RESTful endpoints integrated with Next.js frontends — delivering scalable, low-latency AI-powered APIs.
- **Integrated ElevenLabs and Kokoro LLM** for AI voice synthesis (TTS), enabling real-time, natural human-computer interaction in live applications.
- **Built structured data ingestion pipelines** using Pydantic, Pydub, PyPDF, and Pillow — owning the complete media processing layer.
- **Implemented core LLM integration patterns:** prompt engineering, context window management, and generative AI feature delivery within end-to-end production services.

## Technical Projects

**Cybersecurity Intelligence Platform** — Python — Multi-Agent AI — NVD / CIRCL / MITRE APIs *Oct – Dec 2025*

- **Architected a 4-agent sequential AI pipeline** (threat analysis, CVE research, CVSS scoring, mitigation planning) mirroring production agentic frameworks such as LangGraph and AutoGen.
- **Integrated real-time threat intelligence feeds** (NVD, CIRCL, MITRE ATT&CK); automated markdown reports reduced manual analyst time from ~2 hours to under 6 minutes (**95% reduction**).

**Computer Vision Football Counter** — Python — YOLOv8-nano — ByteTrack — CUDA *Sep – Oct 2025*

- **Deployed YOLOv8-nano + ByteTrack** on a GPU-accelerated pipeline, achieving **79 FPS** on 720p video via deliberate model selection and CUDA memory optimisation.
- **Implemented state-machine bidirectional pedestrian counting**, eliminating double-counts across 3,494 test frames — validated for retail football analytics.

**AI-Powered Knee Osteoarthritis Detection System** — PyTorch — MobileNetV3 — Android Studio *Mar – Jun 2025*

- **Fine-tuned MobileNetV3** for 5-class Kellgren-Lawrence grade classification from knee X-rays (77.6% accuracy). Applied structured pruning + INT8 quantisation to reduce model size by **40%**, enabling sub-second on-device inference on mid-range Android devices.

## Technical Skills

**Languages:** Python, Java

**Backend & APIs:** FastAPI, Uvicorn, Pydantic, REST API Design, Next.js (familiar)

**Tools & Platforms:** Git, GitHub, VS Code, Google Colab, Linux, Android Studio

**Methodologies:** Scrum, Kanban, Jira

## Publications

**LoRA Adapter Weight Tuning with Multi-Task Learning for Faux-Hate Detection** — ACL Anthology — Association for Computational Linguistics *2024*

- Peer-reviewed paper at **ACL 2024**. Proposed a novel LoRA adaptation for parameter-efficient fine-tuning in a multi-task hate speech detection setting — demonstrating research-level expertise in PEFT, NLP, and LLM fine-tuning.

## Certifications

Applied Generative AI — Infosys (2025) — Data Analytics Essentials — Cisco (2025) — Machine Learning with Python — IBM (2024) — Introduction to Java — Sololearn (2025)